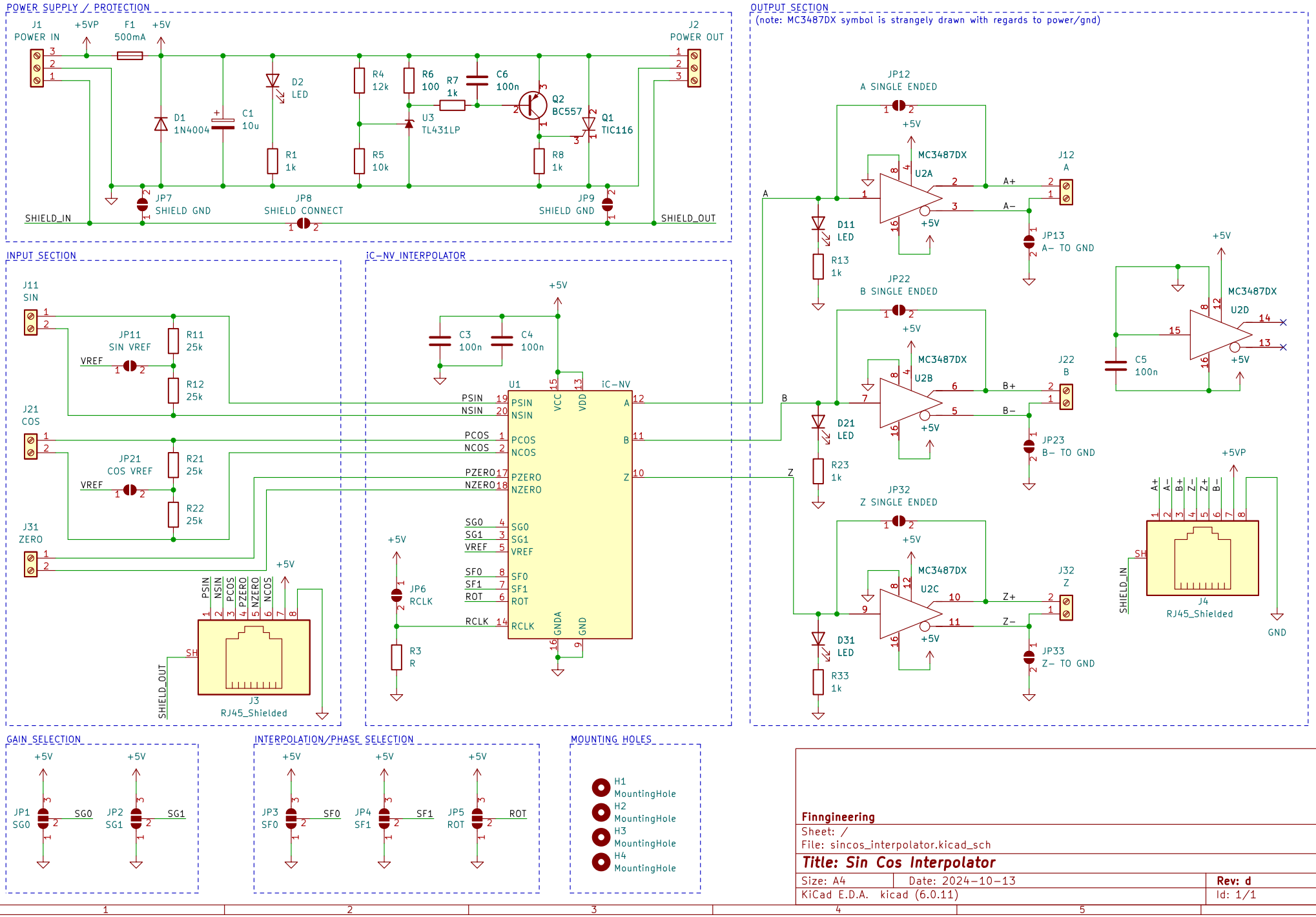

POWER SUPPLY / PROTECTION
J1
POWER IN
+5VP
F1
500mA
+5V
J2
POWER OUT
D1
1N4004
C1
10u
D2
LED
R1
1k
R4
12k
R6
100
R7
1k
C6
100n
U3
TL431LP
Q2
BC557
Q1
TIC116
R5
10k
R8
1k
JP7
SHIELD GND
JP8
SHIELD CONNECT
JP9
SHIELD GND
SHIELD_IN
SHIELD_OUT
INPUT SECTION
J11
SIN
JP11
SIN VREF
VREF
R11
25k
R12
25k
J21
COS
JP21
COS VREF
R21
25k
R22
25k
J31
ZERO
PSIN
NSIN
PCOS
PZERO
NZERO
NCOS
J3
RJ45_Shielded
SH
SHIELD_OUT
iC-NV INTERPOLATOR
U1
iC-NV
C3
100n
C4
100n
VCC
VDD
GNDA
GND
SG0
SG1
VREF
SF0
SF1
ROT
RCLK
JP6
RCLK
R3
R
OUTPUT SECTION
(note: MC3487DX symbol is strangely drawn with regards to power/gnd)
JP12
A SINGLE ENDED
JP22
B SINGLE ENDED
JP32
Z SINGLE ENDED
MC3487DX
U2A
U2B
U2C
U2D
A+
A−
B+
B−
Z+
Z−
J12
A
J22
B
J32
Z
JP13
A− TO GND
JP23
B− TO GND
JP33
Z− TO GND
D11
LED
D21
LED
D31
LED
R13
1k
R23
1k
R33
1k
C5
100n
J4
RJ45_Shielded
SHIELD_IN
GND
GAIN SELECTION
JP1
SG0
JP2
SG1
INTERPOLATION/PHASE SELECTION
JP3
SF0
JP4
SF1
JP5
ROT
MOUNTING HOLES
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Finngineering
Sheet: /
File: sincos_interpolator.kicad_sch
Title: Sin Cos Interpolator
Size: A4
Date: 2024-10-13
Rev: d
KiCad E.D.A. kicad (6.0.11)
Id: 1/1